

The University of Texas at Austin
IN-CLASS WORK 7

Introduction to Financial Mathematics

September 23, 2026

EUROPEAN PUT OPTIONS.

Problem 7.1. (2 points) In which of the following option positions is the investor exposed to an unlimited loss?

- a. Long put option.
- b. Short put option.
- c. Long call option.
- d. Short call option.
- e. None of the above.

Solution. (d)

Just draw the payoff diagrams to convince yourselves.

Problem 7.2. The initial price of the market index is \$900. After 3 months the market index is priced at \$915. The nominal rate of interest convertible monthly is 4.8%. The premium on the put, with a strike price of \$930, is \$8.00. What is the profit at expiration for a **long** put?

- a. \$15.00 loss.
- b. \$6.90 loss.
- c. \$6.90 gain.
- d. \$15.00 gain.
- e. None of the above.

Solution. (c)

The profit from a position is defined as the position's payoff minus the future value of the initial cost.

If $S(T) = 915$ denotes the price of the market index at time $T = 0.25$ (i.e., in three months), then the payoff of the long put is $(K - S(T))_+$, where $K = 930$ denotes the strike of the put. So, since $K > S(T)$, the payoff is

$$(930 - 915)_+ = 15.$$

The future value of the initial put premium is

$$8(1 + 0.004)^3 = 8.0964.$$

So, the profit is

$$15 - 8.0964 = 6.90.$$

Problem 7.3. (3 points) The initial price of the market index is \$1000. After 3 months the market index is priced at \$950. The nominal rate of interest convertible quarterly is 4.0%.

The premium on the long put, with a strike price of \$975, is \$10.00. What is the profit at expiration for this long put?

- a. \$12.00 loss

- b. \$14.90 loss
- c. \$12.00 gain
- d. \$14.90 gain
- e. None of the above.

Solution. (d)

The profit is

$$(K - S(T))_+ - FV_{0,T}[V_P(0)] = (975 - 950)_+ - 10 \left(1 + \frac{0.04}{4} \right) = 25 - 10.10 = 14.90.$$

Problem 7.4. Sample FM(DM) #12

Consider a European put option on a stock index without dividends, with 6 months to expiration, and a strike price of 1,000. Suppose that the nominal annual risk-free rate is 4% **convertible semiannually**, and that the put costs 74.20 now. What price must the index be in 6 months so that being long the put would produce the same profit as being short the put?

- a. \$922.83.
- b. \$924.32.
- c. \$1,000.00.
- d. \$1,075.68.
- e. \$1,077.17.

Solution. (b)

Method I. A quick and insightful way of solving this problem is by realizing that the long-put and the short-put profits are negatives of each other. So, the only way they can be equal is at the “break-even” point. We solve for s in

$$(K - s)_+ - V_P(0) \left(1 + \frac{i^{(2)}}{2} \right) = (1000 - s)_+ - 74.20(1.02) = 0.$$

The solution is $s = 924.32$.

Method II. This is the more pedestrian method. The long-put profit is

$$(K - s)_+ - V_P(0) \left(1 + \frac{i^{(2)}}{2} \right) = (1000 - s)_+ - 74.20(1.02).$$

The short-put profit is the exact negative of the expression above, i.e.,

$$-(K - s)_+ + V_P(0) \left(1 + \frac{i^{(2)}}{2} \right) = -(1000 - s)_+ + 74.20(1.02).$$

So, algebraically, we need to solve for s in the equation

$$\begin{aligned} (1000 - s)_+ - 74.20(1.02) &= -(1000 - s)_+ + 74.20(1.02) \Leftrightarrow 2(1000 - s)_+ = 2 \times 74.20(1.02) \\ &\Leftrightarrow (1000 - s)_+ = 74.20(1.02). \end{aligned}$$

We get the same answer as above, of course.

Problem 7.5. (3 points) *Source: Sample FM(DM) Problem #62.*

The stock price today equals \$100 and its price in one year is modeled by the following distribution:

$$S(1) \sim \begin{cases} 125 & \text{with probability } 1/2 \\ 60 & \text{with probability } 1/2 \end{cases}$$

The annual effective interest rate equals 3%.

Consider an at-the-money, one-year European put option on the above stock whose initial premium is equal to \$7.

What is the expected profit of this put option?

Solution.

$$\frac{1}{2}(100 - 60) - 7(1.03) = 20 - 7.21 = 12.79.$$